# SOBOLEV HOMEOMORPHISM THAT CANNOT BE APPROXIMATED BY DIFFEOMORPHISMS IN $W^{1,1}$

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Abstract. We construct a Sobolev homeomorphism in dimension $n \geq 4$, $f \in W^{1,1}((0,1)^n, \mathbb{R}^n)$ such that $J_f = \det Df > 0$ on a set of positive measure and $J_f < 0$ on a set of positive measure. It follows that there are no diffeomorphisms (or piecewise affine homeomorphisms) $f_k$ such that $f_k \to f$ in $W^{1,1}_{\mathrm{loc}}$.

## 1. Introduction

The general problem of finding suitable approximations of homeomorphisms $f : \mathbb{R}^n \supseteq \Omega \longrightarrow f(\Omega) \subseteq \mathbb{R}^n$ using piecewise affine homeomorphisms has a long history. As far as we know, in the simplest non-trivial setting (i.e. $n = 2$, approximations in the $L^\infty$-norm) the problem was solved by Radó [38]. Due to its fundamental importance in geometric topology, the problem of finding piecewise affine homeomorphic approximations in the $L^\infty$-norm and dimensions $n > 2$ was deeply investigated in the '50s and '60s. In particular, it was solved by Moise [32] and Bing [8] in the case $n = 3$ (see also the survey book [33]), while for contractible spaces of dimension $n \geq 5$ the result follows from theorems of Connell [11], Bing [9], Kirby [25] and Kirby, Siebenmann and Wall [26] (for a proof see, e.g., Rushing [40, Theorem 4.11.1.] or Luukkainen [28]). Finally, twenty years later, while studying the class of quasiconformal manifolds, Donaldson and Sullivan [14] proved that the result is false in dimension 4.

Once completely solved in the uniform sense, the approximation problem suddenly became of interest again in a completely different context, namely for variational models in nonlinear elasticity. Let us briefly explain why. In the setting of nonlinear elasticity (see for instance the pioneering work by Ball [4]), one is led to study existence and regularity properties of minimizers of energy functionals of the form

$$I(f) = \int_\Omega W(Df)\,dx\,, \tag{1.1}$$

where $f : \mathbb{R}^n \supseteq \Omega \to \Delta \subseteq \mathbb{R}^n$ models the deformation of a homogeneous elastic material with respect to a reference configuration $\Omega$ and prescribed boundary values, while $W : \mathbb{R}^{n\times n} \to \mathbb{R}$ is the stored-energy functional. In order for the model to be physically relevant, as pointed out by Ball in [5, 6], one has to require that $u$ is a

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homeomorphism - this corresponds to the non-impenetrability of the material - and that

$$W(A) \to +\infty \quad \text{as } \det A \to 0\,, \qquad W(A) = +\infty \quad \text{if } \det A \leq 0\,. \tag{1.2}$$

The first condition in (1.2) prevents from too high compressions of the elastic body, while the latter guarantees that the orientation is preserved. In particular, if $u$ is an admissible deformation with finite energy, then one has that

$$J_f := \det Df > 0 \quad \text{a.e. in } \Omega \tag{1.3}$$

and hence we restrict our attention to mappings which do not change orientation. Another property of $W$ that appears naturally in many problems of nonlinear elasticity is the *quasiconvexity* (see for instance [3]). Unfortunately, no general existence result is known under condition (1.2), not even if the quasiconvexity assumption is added: one has either to drop condition (1.2) and impose $p$-growth conditions on $W$ (see [36, 1]), or to require that $W$ is *polyconvex* and that some coercivity conditions are satisfied (see [3, 37]). Moreover, also in the cases in which the existence of $W^{1,p}$ minimizers is known, very little is known about their regularity.

As pointed out by Ball in [5, 6] (who ascribes the question to Evans [16]), an important issue toward understanding the regularity of the minimizers in this setting (i.e., $W$ quasiconvex and satisfying (1.2)) would be to show the existence of minimizing sequences given by piecewise affine homeomorphisms or by diffeomorphisms. In particular, a first step would be to prove that any homeomorphism $u \in W^{1,p}(\Omega;\mathbb{R}^n)$, $p \in [1,+\infty)$, can be approximated in $W^{1,p}$ by piecewise affine ones or smooth ones. One of the main reasons why one should want to do that, is that the usual approach for proving regularity is to test the weak equation or the variation formulation by the solution itself; but unfortunately, in general this makes no sense unless some a priori regularity of the solution is known. Therefore, it would be convenient to test the equation with a smooth test mapping in the given class which is close to the given homeomorphism. More generally, a result saying that one can approximate a given homeomorphism with a sequence of smooth (or piecewise affine) homeomorphisms would be extremely useful, because it would significantly simplify many other known proofs, and it would easily lead to stronger new results.

Let us stress that finding diffeomorphisms near a given homeomorphism is not an easy task, as the usual approximation techniques like mollification or Lipschitz extension using the maximal operator destroy, in general, injectivity. And of course we need to approximate our homeomorphism not just with smooth maps, but with smooth homeomorphisms (otherwise the approximating sequence would have nothing to do with the original problem).

Let us now describe the results which are known in the literature in this direction. The first ones were obtained in 2009 by Mora-Corral [34] (for planar bi-Lipschitz mappings that are smooth outside a finite set) and by Bellido and Mora-Corral [7], in which they prove that if $f, f^{-1} \in C^{0,\alpha}$ for some $\alpha \in (0,1]$, then one can find piecewise affine approximations $g$ of $f$ in $C^{0,\beta}$, where $\beta \in (0,\alpha)$ depends only on $\alpha$. Then Iwaniec, Kovalev and Onninen [23] almost completely solved the approximation problem of planar Sobolev homeomorphisms, proving that any homeomorphism $f \in W^{1,p}(\Omega,\mathbb{R}^2)$, for any $1 < p < +\infty$, can be approximated by diffeomorphisms $v$ in the $W^{1,p}$ norm (improving the previous result, of the same authors, for the case $p = 2$,

see [24]). Recently, the missing case of diffeomorphic approximation of $W^{1,1}$ planar homeomorphism was solved by Hencl and Pratelli [22]. Let us also recall the result of Daneri and Pratelli from [12] and [13] that any planar bi-Lipschitz mapping $f$ can be approximated by diffeomorphisms $f_k$ such that $f_k$ converge to $f$ in $W^{1,p}$ norm and simultaneously $f_k^{-1}$ converge to $f^{-1}$ in $W^{1,p}$, giving the first result in which also the distance of the inverse mappings is approximated. For the approximation of mappings that are not homeomorphisms but a limit of homeomorphisms see Iwaniec and Onninen [30] and [31].

As we mentioned before in models of nonlinear elasticity one usually assumes that $J_f > 0$ a.e. (or at least $J_f \geq 0$ a.e.). It is therefore natural to ask if this condition is automatically satisfied (up to a reflection) in the reasonable class of mappings. This problem was promoted by Hajlasz, see e.g. Goldstein and Hajlasz [18]. As each homeomorphism on a domain is either sense-preserving or sense-reversing (see Preliminaries) we can equivalently ask if the topological (sense-preserving) and analytical ($J_f \geq 0$) notion of orientation are the same.

Using a linking number argument it was shown in Hencl and Malý [21] that for Sobolev homeomorphisms $f \in W^{1,1}_{\mathrm{loc}}(\Omega, \mathbb{R}^n)$ on a domain $\Omega$ for $n = 2$ or $n = 3$ we have either $J_f \geq 0$ a.e. in $\Omega$ or $J_f \leq 0$ a.e. in $\Omega$. The same conclusion is known in higher dimension $n \geq 4$ under the assumption $f \in W^{1,p}$ for $p > [n/2]$; here $[.]$ denotes the integer part. One of the aims of this paper is to show that similar result does not hold for $f \in W^{1,1}$ in dimension $n \geq 4$. This solves the open problem promoted by Hajlasz and explicitly mentioned in [27] or later in [20, Open problem 14].

**Theorem 1.1.** *There is a homeomorphism in the Sobolev space $f \in W^{1,1}((-1,1)^4, \mathbb{R}^4)$ such that $\mathcal{L}^4(\{x : \ J_f(x) > 0\}) > 0$ and $\mathcal{L}^4(\{x : \ J_f(x) < 0\}) > 0$.*

In higher dimension we can of course use the mapping

$$F(x_1, \dots, x_n) = [f_1(x_1, x_2, x_3, x_4), \dots, f_4(x_1, x_2, x_3, x_4), x_5, \dots, x_n]$$

to obtain a $W^{1,1}$-Sobolev homeomorphism with similar properties. Let us also note that even for a Sobolev homeomorphism in $W^{1,p}$ for $p < n$ it may happen that $J_f \equiv 0$ a.e. in any dimension $n \geq 2$ - see Hencl [19], Černý [10] and D'Onofrio, Hencl and Schiattarella [15].

Previously we discussed the positive known results of the approximation of Sobolev homeomorphism by diffeomorphisms in the plane, i.e. $n = 2$. Unfortunately no positive or negative results were known in higher dimensions $n \geq 3$. As a simple consequence of our main theorem we obtain the following first result in higher dimension.

**Corollary 1.2.** *Let $n \geq 4$. There exists a homeomorphism in the Sobolev space $f \in W^{1,1}((-1,1)^n, \mathbb{R}^n)$ such that there are no diffeomorphisms (or piecewise affine homeomorphisms) $f_k : (-1,1)^n \to \mathbb{R}^n$ such that $f_k \to f$ in $W^{1,1}_{\mathrm{loc}}((-1,1)^n, \mathbb{R}^n)$.*

Indeed, assume on the contrary that such $f_k$ exist. Then passing to a subsequence we may assume that the pointwise limit satisfies $Df_k(x) \to Df(x)$ for a.e. $x$ and hence $J_{f_k}(x) \to J_f(x)$ for a.e. $x$. As $f_k$ are locally Lipschitz we know that $J_{f_k} \geq 0$ a.e. in $(0,1)^n$ or $J_{f_k} \leq 0$ a.e. in $(0,1)^n$ (see [21] or [20, Theorem 5.22]). Then $J_f$ as a pointwise limit of nonnegative (or nonpositive) functions cannot change sign which gives us the contradiction.

Now we outline the rough idea of our construction without some details. We fix a Cantor type set $C \subset (-1,1)$ of positive measure and we set

$$\begin{aligned}\mathcal{K} :=& \big(C \times C \times C \times [-1,1]\big) \cup \big(C \times C \times [-1,1] \times C\big) \cup \\ &\cup \big(C \times [-1,1] \times C \times C\big) \cup \big([-1,1] \times C \times C \times C\big) \ .\end{aligned}$$

First we construct a sense-preserving homeomorphism $f_1 : \mathbb{R}^n \to \mathbb{R}^n$ such that $f_1([x_1,x_2,x_3,x_4]) = [-x_1,x_2,x_3,x_4]$ for every $x \in \mathcal{K}$ and $f_1$ is locally Lipschitz on $\mathbb{R}^4 \setminus C \times C \times C \times C$. Now we describe how to finish the construction with this $f_1$ and then we describe how to construct such a mapping $f_1$. It is clear that at the points of density of $\mathcal{K}$ we have that the determinant of the approximative derivative of $f_1$ satisfies $J_{f_1} = -1$ and hence the Jacobian does not preserve signs. We need to have a mapping in $W^{1,1}$ and we want to verify the ACL condition (see Preliminaries). The mapping $f_1 : (-1,1)^4 \to \mathbb{R}^4$ is clearly absolutely continuous on $\mathcal{L}^3$-a.e. line parallel to coordinate axes. This is obvious for $z \in C \times C \times C$ as $f_1(t,z) = [-t,z]$ for every $t \in (-1,1)$ there and for $z \notin C \times C \times C$ we know that $f_1(t,z)$ is Lipschitz and hence absolutely continuous. It remains to estimate $\int |Df_1|$ but this is not necessarily finite. We find some nice decomposition of $(-1,1)^3 \setminus C \times C \times C = \bigcup_{i=1}^{\infty} \mathcal{A}_i$, where $\mathcal{A}_i$ are compact, $\mathcal{A}_i \subset \mathcal{A}_{i+1}$ and $\operatorname{dist}(\mathcal{A}_i, C \times C \times C) > 0$ and the following holds: As $f_1$ is Lipschitz on $[-1,1] \times \mathcal{A}_i$ we know that

$$l_i := \sup\big\{\mathcal{H}^1(f_1(L)) : L = (-1,1) \times \{z\} \text{ for some } z \in \mathcal{A}_i\big\} < \infty.$$

As the integral of the derivative corresponds to the length of the image we obtain that the derivative with respect to the first coordinate can be estimated as

$$\begin{aligned}\int_{(-1,1)^4} |D_1 f_1| &= \int_{(-1,1)\times C\times C\times C} |D_1 f_1| + \int_{(-1,1)^4 \setminus ((-1,1)\times C\times C\times C)} |D_1 f_1| \\ &\le \int_{(-1,1)\times C\times C\times C} 1 + \sum_i \mathcal{H}^3\big(\mathcal{A}_i \setminus \mathcal{A}_{i-1}\big) l_i \ ,\end{aligned} \tag{1.4}$$

but the last sum maybe infinite. We thus consider a different Cantor type set $\hat{\mathcal{K}} \subset [-1,1]^4$ of positive measure (possibly larger) homeomorphic to $\mathcal{K}$ such that for the corresponding sets $\hat{\mathcal{A}}_i$ (possibly much smaller) we have $\sum_i \mathcal{L}^4(\hat{\mathcal{A}}_i \setminus \hat{\mathcal{A}}_{i-1}) l_i < \infty$. We transfer $f_1$ which turns over $\mathcal{K}$ to the mapping $f$ which turns over $\hat{\mathcal{K}}$ and for which we can use analogy of (1.4) (and analogous estimates for derivatives with respect to other coordinates) to finish the proof of $f \in W^{1,1}$. Other properties like the sign of the Jacobian are transfered easily as well.

To construct the homeomorphism $f_1$ as above we first realize that it is enough to do the construction for Cantor type set $C_0 \subset (-1,1)$ of dimension $0 < \delta < 1/6$ (instead of $C$ of positive measure) as we can easily map the corresponding $\mathcal{K}_0$ (of dimension $1 + 3\delta$) to $\mathcal{K}$ by a sense-preserving homeomorphism that is locally bi-Lipschitz outside of this Cantor type set. It is possible to find a direction in $\mathbb{R}^4$ such that the projection of $\mathcal{K}_0$ to the corresponding hyperplane is one-to-one. The rough reason for that is that the set of directions where the projection is not one-to-one has Hausdorff dimension at most $\dim \mathcal{K}_0 + \dim \mathcal{K}_0 = 2 + 6\delta$ (starting+ending point of the direction) and this is smaller than 3-the dimension of all directions. This projection of $\mathcal{K}_0$ can be extended to the homeomorphism $g : \mathbb{R}^4 \to \mathbb{R}^4$ which is bi-Lipschitz. With some additional work we can moreover require that the image of $\mathcal{K}_0$ on the

hyperplane is symmetric and by the turnover of the 3-dimensional hyperplane with respect to one direction (which can be done by a sense-preserving homeomorphism of $\mathbb{R}^4$) and the composition with $g^{-1}$ we obtain our mapping $f_1$. By the symmetry of the image of $\mathcal{K}_0$ on the hyperplane and the turnover of the hyperplane we obtain the key property $f_1(x_1, x_2, x_3, x_4) = [-x_1, x_2, x_3, x_4]$ for every $x \in \mathcal{K}_0$.

Let us briefly discuss the sharpness of our results for example in dimension $n = 4$. It was shown in [21] that the Jacobian cannot change sign for homeomorphisms in $W^{1,p}((0,1)^4, \mathbb{R}^4)$, $p > 2$. Therefore at the moment we do not know the result or the counterexample for $1 < p \leq 2$. We are not aware of any topological objection that would prevent the existence of such a counterexample. On the other hand our construction is not entirely explicit and hence cannot be used easily for the estimate of $\int |Df|^p$, $1 < p \leq 2$. In particular we show that there is some direction for which the projection of our Cantor type set $\mathcal{K}_0$ is one-to-one as the dimension of the bad directions is too small. Then we show that there is some extension and at the end we somehow (again not explicitly) increase the size of the Cantor type set to decrease the measure of the set where $|Df|$ is too big.

## 2. Preliminaries

By $B(c, r)$ we denote the ball centered at $c \in \mathbb{R}^n$ with radius $r > 0$ and by $Q(c, r)$ we denote the corresponding cube. A point $x \in \mathbb{R}^n$ is denoted in coordinates as $[x_1, x_2, \ldots, x_n]$.

2.1. **ACL condition.** It is a well-known fact (see e.g. [2, Section 3.11]) that a mapping $u \in L^1_{\mathrm{loc}}(\Omega, \mathbb{R}^m)$ is in $W^{1,1}_{\mathrm{loc}}(\Omega, \mathbb{R}^m)$ if and only if there is a representative which is an absolutely continuous function on almost all lines parallel to coordinate axes and the derivative on these lines is integrable. More precisely, let $i \in \{1, 2, \ldots, n\}$ and denote by $\pi_i$ the projection on the hyperplane perpendicular to the $x_i$-axis. Suppose that $Q(c, r) := (c_1 - r, c_1 + r) \times \ldots \times (c_n - r, c_n + r) \subset \Omega$ for some $c \in \mathbb{R}^n$, $r > 0$ and set $Q^i(c, r) = \pi_i(Q(c, r))$. Let $y \in Q^i(c, r)$ and denote

$$u_{i,y}(t) = u(y_1, \ldots, y_{i-1}, y_i + t, y_{i+1}, \ldots, y_n) \quad \text{for} \quad t \in (c_i - r, c_i + r).$$

**Theorem 2.1.** *Let $\Omega \subset \mathbb{R}^n$ be open and let $u \in L^1_{\mathrm{loc}}(\Omega, \mathbb{R}^m)$. Then $u \in W^{1,1}_{\mathrm{loc}}(\Omega, \mathbb{R}^m)$ if and only if the following happens. For every cube $Q(c, r) \subset\subset \Omega$ and for every $i \in \{1, \ldots, n\}$ there is a representative $\tilde{u}$ of $u$ such that the function $\tilde{u}_{i,y}(t)$ is absolutely continuous on $(c_i - r, c_i + r)$ (i.e. each coordinate function is absolutely continuous) for $\mathcal{L}^{n-1}$ almost every $y \in Q^i(c, r)$ and moreover*

$$\int_{Q^i(c,r)} \int_{c_i - r}^{c_i + r} |\nabla \tilde{u}_{i,y}(t)| \, dt \, dy < \infty.$$

**Topological degree.** Given a smooth map $f$ from $\Omega \subset \mathbb{R}^n$ into $\mathbb{R}^n$ we can define the topological degree as

$$\deg(f, \Omega, y_0) = \sum_{\{x \in \Omega : f(x) = y_0\}} \operatorname{sgn}(J_f(x))$$

if $J_f(x) \neq 0$ for each $x \in f^{-1}(y_0)$. This definition can be extended to arbitrary continuous mappings and each point, see e.g. [17].

A continuous mapping $f : \Omega \to \mathbb{R}^n$ is called *sense-preserving* if

$$\deg(f, \Omega', y_0) > 0$$

for all domains $\Omega' \subset\subset \Omega$ and all $y_0 \in f(\Omega') \setminus f(\partial\Omega')$. Similarly we call $f$ *sense-reversing* if $\deg(f, \Omega', y_0) < 0$ for all $\Omega'$ and $y_0$. Let us recall that each homeomorphism on a domain is either sense-preserving or sense-reversing see [39, II.2.4., Theorem 3].

2.2. **Hausdorff dimension.** Let $\alpha > 0$ and $\varepsilon > 0$. We use the usual Hausdorff measure of a set $E \subset \mathbb{R}^n$, i.e.

$$\mathcal{H}^\alpha_\varepsilon(E) = \inf\Big\{\sum_{i=1}^{\infty} \operatorname{diam}^\alpha A_i : \ E \subset \bigcup_{i=1}^{\infty} A_i, \ \operatorname{diam} A_i < \varepsilon\Big\} \text{ and } \mathcal{H}^\alpha(E) = \lim_{\varepsilon\to 0+} \mathcal{H}^\alpha_\varepsilon(E) \ .$$

The Hausdorff dimension of a set $E$ is

$$\dim_{\mathcal{H}}(E) = \sup\{\alpha > 0 : \ \mathcal{H}^\alpha(E) = \infty\} = \inf\{\alpha > 0 : \ \mathcal{H}^\alpha(E) = 0\} \ .$$

It is well-known fact that Lipschitz mappings do not raise the Hausdorff dimension and it is also easy to see that countable union of sets does not raise the Hausdorff dimension.

## 3. Sense preserving homeomorphism that turns over $\mathcal{K}_0$

**Lemma 3.1.** *Let $M \subseteq \mathbb{R}^n$ be a compact set whose Hausdorff dimension is less than $\frac{n-1}{2}$. Then for every $\varepsilon > 0$ and for every direction $u \in \mathbb{R}^n$ there is a direction $v \in \mathbb{R}^n$ such that $\|u - v\| < \varepsilon$ and every line with direction $v$ intersects $M$ in at most one point.*

*Proof.* Consider a mapping $\varphi\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ defined by $\varphi(x, y) = x - y$. Clearly such a map $\varphi$ is Lipschitz and thus it does not raise the Hausdorff dimension. Moreover, the Hausdorff dimension of $M \times M$ is less than $\frac{n-1}{2} + \frac{n-1}{2} = n - 1$ and it follows that the Hausdorff dimension of $\varphi(M \times M)$ is less than $n - 1$.

Let us denote $N = \varphi(M \times M) \setminus \{0\}$ and by $\pi\colon \mathbb{R}^n \setminus \{0\} \to S^{n-1}(0, 1)$ the radial projection, i.e. the map satisfying $\pi(v) = \frac{v}{\|v\|}$. Clearly $\pi|_{\mathbb{R}^n \setminus B(0,\frac{1}{m})}$ is Lipschitz for a fixed $m \in \mathbb{N}$ and thus the Hausdorff dimension of

$$\pi(N) = \pi\big(\bigcup_m (N \setminus B(0, \tfrac{1}{m}))\big) = \bigcup_m \pi\big(N \setminus B(0, \tfrac{1}{m})\big)$$

is less than $n - 1$ because the countable union does not raise the dimension. Since the Hausdorff dimension of any nonempty open subset of $S^{n-1}$ is equal to $n-1$, there exists $v \in S^{n-1} \setminus \pi(N)$ such that $\|u - v\| < \varepsilon$. □

In the following lemma we observe that some special self similar sets are in fact graphs of Lipschitz functions.

**Lemma 3.2.** *Let $M \subseteq \mathbb{R}^4$ be a compact set. Suppose there exists $\eta > 0$ such that for every pair of points $a, b \in M$ there are points $c, d \in M$ such that $\|c - d\| > \eta$ and the vectors $a - b$ and $c - d$ are parallel. Suppose moreover that for every point $x \in \pi_{123}(M)$ there is just one $t(x) \in \mathbb{R}$ such that $[x, t(x)] \in M$, where $\pi_{123}$ denotes the projection to the first three coordinates. Then $t$ is a Lipschitz function from $\pi_{123}(M)$ to $\mathbb{R}$.*

*Proof.* Suppose for contradiction that $t$ is not a Lipschitz function. Thus for every $n \in \mathbb{N}$ there is a pair of points $a_n, b_n$ in $\mathbb{R}^3$ such that $|t(a_n) - t(b_n)| > n\|a_n - b_n\|$. By the assumption applied to points $A_n = [a_n, t(a_n)], B_n = [b_n, t(b_n)]$ there are points $C_n = [c_n, t(c_n)]$ and $D_n = [d_n, t(d_n)]$ such that $A_n - B_n$ and $C_n - D_n$ are parallel and $\|C_n - D_n\| > \eta$.

Suppose for a while that $\liminf \|c_n - d_n\| = 0$. Since $t$ is continuous and thus also uniformly continuous (as its domain is a compact set) it follows that $\liminf \|t(c_n) - t(d_n)\| = 0$. Hence $\liminf \|[c_n, t(c_n)] - [d_n, t(d_n)]\| = 0$ which is a contradiction since $\eta > 0$. Hence there is some $\delta > 0$ such that $\|c_n - d_n\| > \delta$ for $n \in \mathbb{N}$. Since the vectors $A_n - B_n$ and $C_n - D_n$ are parallel it follows that

$$|t(c_n) - t(d_n)| = \frac{|t(a_n) - t(b_n)|}{\|a_n - b_n\|} \|c_n - d_n\| > n\delta \to \infty.$$

This is a contradiction, since the function $t$ is bounded. □

We need to define Cantor set $C_0 \subset [-1, 1]$ of Hausdorff dimension $\delta := 1/7$. By $\mathbb{V}$ we denote the set of $2^4$ vertices of the cube $[-1, 1]^4$. The sets $\mathbb{V}^k = \mathbb{V} \times \ldots \times \mathbb{V}$, $k \in \mathbb{N}$, will serve as the sets of indices for our construction. Set $z_0 = \tilde{z}_0 = 0$ and let us define

$$\tilde{a}_k = 2^{-6k} \text{ and } \tilde{r}_k = \tilde{a}_k 2^{-k} .$$

It follows that $(-1, 1)^4 = Q(z_0, \tilde{r}_0)$ and further we proceed by induction. For $\boldsymbol{v} = [v_1, \ldots, v_k] \in \mathbb{V}^k$ we denote $\boldsymbol{w} = [v_1, \ldots, v_{k-1}]$ and we define

$$\tilde{z}_{\boldsymbol{v}} = \tilde{z}_{\boldsymbol{w}} + \frac{1}{2}\tilde{r}_{k-1} v_k = z_0 + \frac{1}{2}\sum_{j=1}^{k} \tilde{r}_{j-1} v_j,$$

$$\tilde{Q}'_{\boldsymbol{v}} = Q(\tilde{z}_{\boldsymbol{v}}, \tfrac{\tilde{r}_{k-1}}{2}) \text{ and } \tilde{Q}_{\boldsymbol{v}} = Q(\tilde{z}_{\boldsymbol{v}}, \tilde{r}_k).$$

Formally we should write $\boldsymbol{w}(\boldsymbol{v})$ instead of $\boldsymbol{w}$ but for the simplification of the notation we will avoid this.

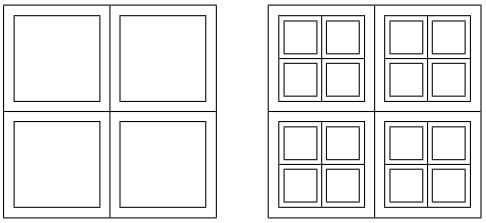


**Fig. 1.** Two-dimensional projection of cubes $\tilde{Q}_{\boldsymbol{v}}$ and $\tilde{Q}'_{\boldsymbol{v}}$ for $\boldsymbol{v} \in \mathbb{V}^1$ and $\boldsymbol{v} \in \mathbb{V}^2$.

The number of the cubes $\{\tilde{Q}_{\boldsymbol{v}} : \boldsymbol{v} \in \mathbb{V}^k\}$ is $2^{4k}$. It is not difficult to find out that the resulting Cantor set

$$\bigcap_{k=1}^{\infty} \bigcup_{\boldsymbol{v} \in \mathbb{V}^k} \tilde{Q}_{\boldsymbol{v}} =: \mathcal{C}_0 = C_0 \times C_0 \times C_0 \times C_0$$

is a product of 4 Cantor sets $C_0$ in $\mathbb{R}$. Moreover, $\mathcal{L}^4(\mathcal{C}_0) = 0$ since

$$\mathcal{L}^4\Big(\bigcup_{\boldsymbol{v} \in \mathbb{V}^k} \tilde{Q}_{\boldsymbol{v}}\Big) = 2^{4k}(2\tilde{a}_k 2^{-k})^4 \overset{k \to \infty}{\to} 0.$$

We can easily find out that the Hausdorff dimension of $C_0$ is $1/7$ as in the $k$-th step of its construction we have $2^k$ intervals of length $2\tilde{a}_k 2^{-k} = 2 \cdot 2^{-7k}$. We note that $C_0$ is symmetrical, i.e. $-C_0 = C_0$ and that $(-\frac{1}{4}, \frac{1}{4}) \cap C_0 = \emptyset$.

Let us set

$$\begin{aligned}\mathcal{K}_0 =&(C_0 \times C_0 \times C_0 \times [-1,1]) \cup (C_0 \times C_0 \times [-1,1] \times C_0) \cup \\ &\cup (C_0 \times [-1,1] \times C_0 \times C_0) \cup ([-1,1] \times C_0 \times C_0 \times C_0).\end{aligned} \tag{3.1}$$

We denote by $H$ the hyperplane $\mathbb{R}^3 \times \{0\}$ and by $s\colon \mathbb{R}^4 \to \mathbb{R}^4$ the symmetry with respect to the hyperplane $\{0\} \times \mathbb{R}^3$, i.e. a mapping which satisfies $s(a,b,c,d) = [-a,b,c,d]$.

**Proposition 3.3.** *There is a sense preserving bi-Lipschitz homeomorphism $h\colon \mathbb{R}^4 \to \mathbb{R}^4$ such that $h(\mathcal{K}_0) \subseteq H$ and for every $x \in \mathcal{K}_0$ it holds that $h(s(x)) = s(h(x))$.*

*Proof.* Let us call a vector $v \in \mathbb{R}^4$ good for a set $M \subseteq \mathbb{R}^4$ if every line with direction $v$ intersects $M$ in at most one point. Clearly $(-2\eta, 2\eta) \cap C_0 = \emptyset$ for $\eta = 1/8$ and the Hausdorff dimension of $\mathcal{K}_0$ is $1 + 3\delta < \frac{3}{2}$ (recall that $\delta = \frac{1}{7}$ is the dimension of $C_0$). Hence by Lemma 3.1 there is a good vector $v \in \mathbb{R}^4$ for $\mathcal{K}_0$ such that $\left|\frac{v_1}{v_4}\right| < \frac{\eta}{4}$.

*Step 1. Symmetrical deformation of $\mathcal{K}_0$ so that $[0, v_2, v_3, v_4]$ is a good vector.*

Let us denote $v' = [-v_1, v_2, v_3, v_4] = s(v)$. In what follows we define a homeomorphism $h_1$ of $\mathbb{R}^4$ which will be symmetrical in the first coordinate and such that $[0, v_2, v_3, v_4]$ is a good vector for $h_1(\mathcal{K}_0)$. Such a homeomorphism can be realized by pushing the points only in the direction of $x_1$ axes such that every segment in the region $A$ (see Fig.2) (respectively $A' := s(A)$) with direction $v$ (respectively $v'$) moves to a segment with direction $[0, v_2, v_3, v_4]$. The broken lines $L$ and $L'$ in Figure 2 move to the corresponding vertical dotted lines and all the points for which $x_1 = 0$ are fixed.

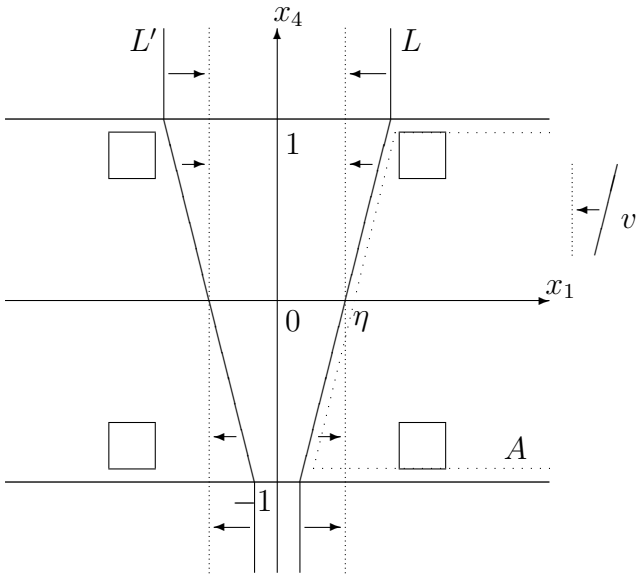



**Fig. 2.** The homeomorphism $h_1$.

This can be formally defined for $x_1 \geq 0$ as

$$h_1(x) = \begin{cases} [x_1 - x_4\frac{v_1}{v_4}, x_2, x_3, x_4] & \text{for } x_4 \in [-1,1], x_4 \geq (x_1-\eta)\frac{v_1}{v_4} \\ [x_1\frac{\eta}{\eta+\frac{v_1}{v_4}x_4}, x_2, x_3, x_4] & \text{for } x_4 \in [-1,1], x_4 < (x_1-\eta)\frac{v_1}{v_4} \\ [x_1 - \frac{v_1}{v_4}, x_2, x_3, x_4] & \text{for } x_4 > 1, x_1 \geq \eta + \frac{v_1}{v_4} \\ [x_1\frac{\eta}{\eta+\frac{v_1}{v_4}}, x_2, x_3, x_4] & \text{for } x_4 > 1, x_1 < \eta + \frac{v_1}{v_4} \\ [x_1 + \frac{v_1}{v_4}, x_2, x_3, x_4] & \text{for } x_4 < -1, x_1 \geq \eta - \frac{v_1}{v_4} \\ [x_1\frac{\eta}{\eta-\frac{v_1}{v_4}}, x_2, x_3, x_4] & \text{for } x_4 < -1, x_1 < \eta - \frac{v_1}{v_4} \end{cases}$$

and for $x_1 < 0$ we define symmetrically $h_1(x) = s(h_1(s(x)))$.

As $|\frac{v_1}{v_4}| < \frac{\eta}{4}$ it can be easily verified that $h_1$ is a bi-Lipschitz homeomorphism of $\mathbb{R}^4$, such that $h_1 \circ s = s \circ h_1$. We will verify that $[0, v_2, v_3, v_4]$ is a good vector for $h_1(\mathcal{K}_0)$. Let $M$ be a segment in the strip $|x_4| \leq 1$ with such a direction. This can be assumed since $h_1(\mathcal{K}_0)$ is a subset of this strip. Clearly the first coordinates of points in $M$ are all the same. Suppose first that this coordinate is at least $\eta$ in the absolute value. It follows that $h_1^{-1}(M)$ is a segment with direction $v$ or $v'$ and thus it intersects $\mathcal{K}_0$ in at most one point. Hence the segment $M$ intersects $h_1(\mathcal{K}_0)$ in at most one point as well. Suppose now that the first coordinate of points in $M$ is less than $\eta$ in the absolute value. Then $h_1^{-1}(M)$ does not intersect $C_0 \times [-1,1]^3$ and hence

$$h_1(\mathcal{K}_0) \cap ((-\eta, \eta) \times \mathbb{R}^3)) = (-\eta, \eta) \times C_0^3.$$

Since $h_1$ is changing the first coordinate only it follows that $M$ intersects $\mathcal{K}_0$ in at most one point. Consequently $[0, v_2, v_3, v_4]$ is a good vector for $h_1(\mathcal{K}_0)$.

*Step 2. Deformation of $\mathcal{K}_0$ so that $[0,0,0,1]$ is a good vector.*

Let us define a linear homeomorphism

$$h_2(x_1, x_2, x_3, x_4) = \left[x_1, x_2 - x_4\frac{v_2}{v_4}, x_3 - x_4\frac{v_3}{v_4}, \frac{x_4}{v_4}\right], x \in \mathbb{R}^4.$$

It follows that $h_2$ moves every line with direction $[0, v_2, v_3, v_4]$ onto a line with direction $[0,0,0,1]$ and that $h_2 \circ s = s \circ h_2$. Since the vector $[0, v_2, v_3, v_4]$ is good for $h_1(\mathcal{K}_0)$ it follows that the vector $[0,0,0,1]$ is good for $h_2(h_1(\mathcal{K}_0))$.

*Step 3. Lipschitz property on parts of $\mathcal{K}_0$*

Denote by $\pi_{123}$ the projection onto the first three coordinates and let

$$D = \pi_{123}(h_2(h_1(\mathcal{K}_0))).$$

Since $[0,0,0,1]$ is a good vector for $h_2(h_1(\mathcal{K}_0))$ there is just one $t(z) \in \mathbb{R}$ for each $z \in D$ such that $[z, t(z)] \in h_2(h_1(\mathcal{K}_0))$. We can easily verify that $t$ is even in the first coordinate, because $[z, t(z)] \in h_2(h_1(\mathcal{K}_0))$ and thus

$$s(z, t(z)) \in s\big(h_2(h_1(\mathcal{K}_0))\big) = h_2\big(h_1(s(\mathcal{K}_0))\big) = h_2(h_1(\mathcal{K}_0))$$

and so $[-z_1, z_2, z_3, t(z)] \in h_2(h_1(\mathcal{K}_0))$ which means that $t(-z_1, z_2, z_3) = t(z_1, z_2, z_3)$.

We need to verify that $t$ is Lipschitz. Let

$$H_i = h_2\big(h_1(C_0^{i-1} \times [-1,1] \times C_0^{4-i})\big),\ D_i = \pi_{123}(H_i) \text{ for } i = 1,2,3,4.$$

The set $C_0$ is self similar and hence we can find out that the set $H_1$ is self similar and we can divide $H_i$, $i = 2,3,4$, into two parts that are self similar ($H_i \cap \{x_1 < 0\}$

and $H_i \cap \{x_1 > 0\}$). It follows that the assumptions of Lemma 3.2 are satisfied when applied to the set $H_i$ and hence that $t|D_i$ is Lipschitz for every $i \leq 4$.

*Step 4. Lipschitz property on the whole $\mathcal{K}_0$*

It remains to show that $t$ is Lipschitz on the whole union $D = D_1 \cup D_2 \cup D_3 \cup D_4$. We show equivalently that the mapping $x \mapsto (x, t(x))$ is Lipschitz on $D$. To show this it is enough to verify that the map $g : x \mapsto h_1^{-1}(h_2^{-1}(x, t(x)))$ is Lipschitz since $h_1$ and $h_2$ are bi-Lipschitz. By the previous step we know that each of the maps $t|D_i$ is Lipschitz and thus $g|D_i$ is Lipschitz as well. Let us denote by $c_i$ the corresponding Lipschitz constants.

Note that by the definition of $h_1$ and $h_2$ every segment contained in $\mathcal{K}_0$ is moved by $h_2 \circ h_1$ onto another segment. Moreover, if we have two segments in $\mathcal{K}_0$ which are parallel with $x_1, x_2$ or $x_3$ axes, they are moved by $h_2 \circ h_1$ onto parallel segments as well. The only technical detail is that two segments in $\mathcal{K}_0$ parallel with $x_4$ axes can be moved onto segments which are not parallel depending if $x_1 > 0$ or $x_1 < 0$. In this case there is always the same angle between them.

Let us denote by $u_i$ the directions of images under $h_2 \circ h_1$ of the segments in $\mathcal{K}_0$ parallel to $x_i$ axes for $i = 1, 2, 3$. Let $u_4$ be one of the two distinct directions of images under $h_2 \circ h_1$ of the segments in $\mathcal{K}_0$ parallel to $x_4$ axes. Let us denote by $\varphi$ the minimal angle in $[0, \pi/2]$ between $u_i$ and the vector span $\langle u_j, u_k, u_l \rangle$ where $\{i, j, k, l\} = \{1, 2, 3, 4\}$. Note that all of these angles are nonzero because by the definition of $h_2 \circ h_1$ the vectors $u_1$, $u_2$, $u_3$, $u_4$ are linearly independent.

Suppose that $a, e \in D$ are given. Suppose for example that $a \in D_1$ and $e \in D_4$ and denote $g(a) = [x_1, x_2, x_3, x_4]$, $g(e) = [y_1, y_2, y_3, y_4]$. There are points $b \in D_1 \cap D_2$, $c \in D_2 \cap D_3$ and $d \in D_3 \cap D_4$ such that the segments $ab$, $bc$, $cd$ and $de$ are parallel to the vectors $u_1, u_2, u_3$ and $u_4$ (or $s(u_4)$) respectively. For example $b, c, d$ can be chosen so that

$$\begin{aligned} g(b) &= [y_1, x_2, x_3, x_4] \\ g(c) &= [y_1, y_2, x_3, x_4] \\ g(d) &= [y_1, y_2, y_3, x_4]. \end{aligned}$$

It holds that $\|a-c\| \geq \sin\varphi \|a-b\|$, $\|a-d\| \geq \sin\varphi \|a-c\|$, $\|a-e\| \geq \sin\varphi \|a-d\|$, hence $\|a-e\| \geq \sin^3\varphi \|a-b\|$ which implies

$$\|a-b\| \leq \frac{\|a-e\|}{\sin^3\varphi}.$$

The same upper estimate holds for $\|b-c\|$, $\|c-d\|$ and $\|d-e\|$. It follows that

$$\begin{aligned} \|g(a)-g(e)\| &\leq \|g(a)-g(b)\| + ... + \|g(d)-g(e)\| \\ &\leq c_1\|a-b\| + ... + c_4\|d-e\| \\ &\leq (c_1+c_2+c_3+c_4)\frac{1}{\sin^3\varphi}\|a-e\|. \end{aligned}$$

Thus the mapping $g$ is Lipschitz.

*Step 5. Squeezing $\mathcal{K}_0$ into the hyperplane*

Since the function $t\colon D \to \mathbb{R}$ is Lipschitz it can be extended to a function $\bar{t}\colon \mathbb{R}^3 \to \mathbb{R}$ with the same Lipschitz constant. Let us denote

$$h_3(x) = [x_1, x_2, x_3, x_4 - \bar{t}(x_1, x_2, x_3)], x \in \mathbb{R}^4.$$

Clearly $h_3(s(x)) = s(h_3(x))$ for $x \in h_2(h_1(\mathcal{K}_0))$, because $t$ is even in the first coordinate. Since the function $\bar{t}$ is Lipschitz it follows that $h_3$ is Lipschitz too. Note that the inverse of $h_3$ is given by the simple formula

$$h_3^{-1}(x_1, x_2, x_3, x_4) = [x_1, x_2, x_3, x_4 + \bar{t}(x_1, x_2, x_3)], x \in \mathbb{R}^4$$

and thus it is Lipschitz as well. It follows that $h_3$ is a bi-Lipschitz homeomorphism.

Let $h = h_3 \circ h_2 \circ h_1$. Clearly $h(\mathcal{K}_0) \subseteq H$ since $h(\mathcal{K}_0) = h_3(h_2(h_1(\mathcal{K}_0))) = h_3(\text{graph}f) = D \times \{0\}$ and $h \circ s = s \circ h$ on $\mathcal{K}_0$ as $h_i \circ s = s \circ h_i$ on $\mathcal{K}_0$ for $i = 1, 2, 3$. Moreover $h$ being a composition of three bi-Lipschitz mappings is bi-Lipschitz as well. By the same reason $h$ is sense preserving. □

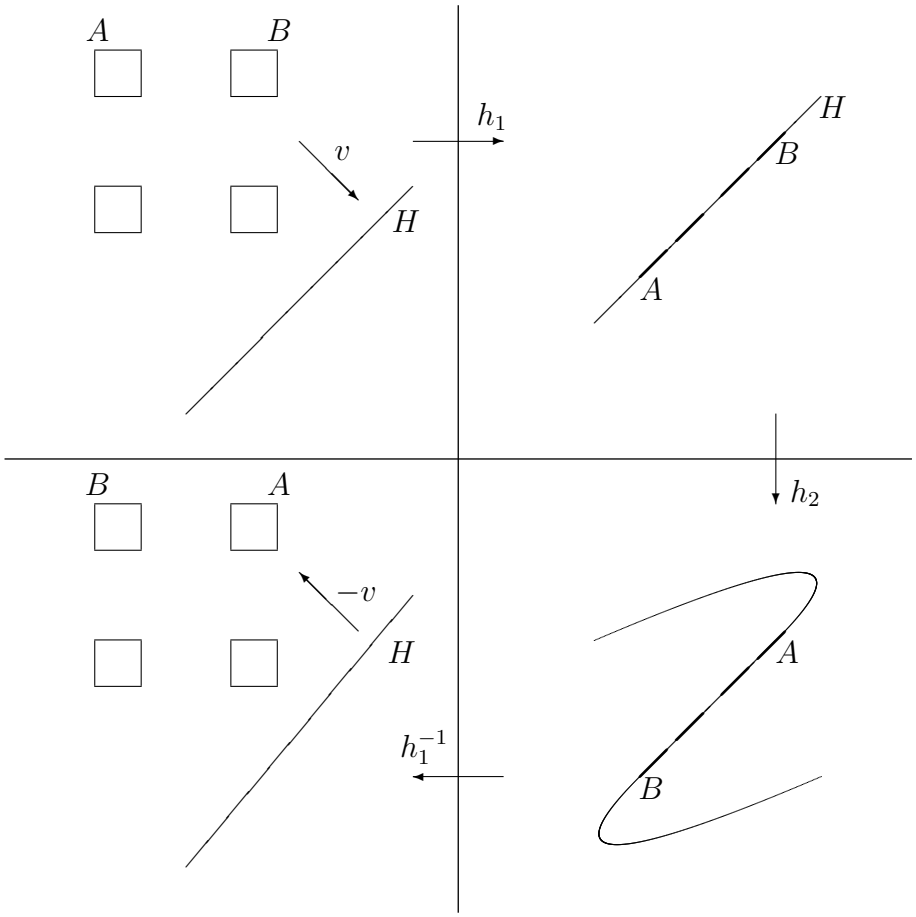



**Fig. 3.** Projecting, turning over and returning back.

**Theorem 3.4.** *There is a sense preserving bi-Lipschitz homeomorphism* $h\colon \mathbb{R}^4 \to \mathbb{R}^4$, *such that*

$$h(x_1, x_2, x_3, x_4) = [-x_1, x_2, x_3, x_4] \text{ for every } x \in \mathcal{K}_0. \tag{3.2}$$

*Proof.* The informal idea of the proof is captured in Figure 3. Let us consider the homeomorphism given by Proposition 3.3, i.e. a bi-Lipschitz homeomorphism $h_1\colon \mathbb{R}^4 \to \mathbb{R}^4$ such that $h_1(\mathcal{K}_0) \subseteq H$ and for which $h_1 \circ s = s \circ h_1$. It is easy to see that there is a sense preserving bi-Lipschitz diffeomorphism $h_2\colon \mathbb{R}^4 \to \mathbb{R}^4$ such that $h_2(x) = s(x)$ for every $x \in H$, $\|x\| \leq \max\{\|h_1(y)\|\colon y \in \mathcal{K}_0\}$. It is enough to define $h = h_1^{-1} \circ h_2 \circ h_1$. It follows that $h$ is a bi-Lipschitz homeomorphism. Moreover for

every $x \in \mathcal{K}_0$ it holds that

$$h(x) = h_1^{-1} \circ h_2 \circ h_1(x) = h_1^{-1} \circ s \circ h_1(x) = h_1^{-1} \circ h_1(s(x)) = s(x).$$

□

## 4. Proof of Theorem 1.1

*Step 1. Construction of Cantor type sets $C$ and $\mathcal{K}$ of positive measure.*

Analogously to the previous section we need to define a Cantor type set $C \subset [-1,1]$ and we use the same notation for $\mathbb{V}^k$ and $\boldsymbol{v} \in \mathbb{V}^k$. Set $z_0 = \tilde{z}_0 = 0$ and let us define

$$a_k = \frac{1}{2}\Big(1 + \frac{1}{k+1}\Big) \quad \text{and} \quad r_k = a_k 2^{-k} .$$

As before we define

$$z_{\boldsymbol{v}} = z_{\boldsymbol{w}} + \frac{1}{2} r_{k-1} v_k = z_0 + \frac{1}{2} \sum_{j=1}^{k} r_{j-1} v_j,$$

$$Q'_{\boldsymbol{v}} = Q(z_{\boldsymbol{v}}, \tfrac{r_{k-1}}{2}) \text{ and } Q_{\boldsymbol{v}} = Q(z_{\boldsymbol{v}}, r_k).$$

It is not difficult to find out that the resulting Cantor set

$$\bigcap_{k=1}^{\infty} \bigcup_{\boldsymbol{v} \in \mathbb{V}^k} Q_{\boldsymbol{v}} =: \mathcal{C} = C \times C \times C \times C$$

is a product of 4 Cantor sets $C$ in $\mathbb{R}$. Moreover, $\mathcal{L}^4(\mathcal{C}) > 0$ since

$$\mathcal{L}^4\Big( \bigcup_{\boldsymbol{v} \in \mathbb{V}^k} Q_{\boldsymbol{v}}\Big) = 2^{4k} (2 a_k 2^{-k})^4 \overset{k \to \infty}{\to} 1.$$

As pointed out in the introduction we need to find a homeomorphism which turns over the set

$$\begin{aligned} \mathcal{K} :=& \big(C \times C \times C \times [-1,1]\big) \cup \big(C \times C \times [-1,1] \times C\big) \cup \\ & \cup \big(C \times [-1,1] \times C \times C\big) \cup \big([-1,1] \times C \times C \times C\big) . \end{aligned} \tag{4.1}$$

For that we would like to use homeomorphism $h$ from Theorem 3.4 which turns over a different set $\mathcal{K}_0$. Let us recall that in the previous section in (3.1) we have defined $\mathcal{K}_0$ using

$$\tilde{a}_k = 2^{-6k} \text{ and } \tilde{r}_k = \tilde{a}_k 2^{-k} .$$

*Step 2. Homeomorphism that maps $\mathcal{K}$ onto $\mathcal{K}_0$.*

Let us denote by $\pi([x_1, x_2, x_3, x_4]) = x_1$ the projection to the first coordinate. It is easy to see that the Cantor type sets $C \subset [-1,1]$ and $C_0 \subset [-1,1]$ are homeomorphic. In particular there is a homeomorphism $t : \mathbb{R} \to \mathbb{R}$ which maps $C$ onto $C_0$. It is just enough to consider a mapping which maps each of the two parts of $\pi(Q'_{\boldsymbol{v}}) \setminus \pi(Q_{\boldsymbol{v}})$ onto the corresponding part of $\pi(\tilde{Q}'_{\boldsymbol{v}}) \setminus \pi(\tilde{Q}_{\boldsymbol{v}})$ linearly for each $\boldsymbol{v} \in \mathbb{V}^k$ and $k \in \mathbb{N}$.

In dimension 4 we consider a mapping

$$T(x_1, x_2, x_3, x_4) = [t(x_1), t(x_2), t(x_3), t(x_4)] .$$

It is easy to check that this mapping is a homeomorphism, maps $\mathcal{K}$ onto $\mathcal{K}_0$ and moreover it maps every line segment parallel to coordinate axes to segment parallel to coordinate axes.

The mapping $h$ from Theorem 3.4 turns over the set $\mathcal{K}_0$ and hence the mapping $h \circ T$ maps $\mathcal{K}$ to $\mathcal{K}_0$ and then turns $\mathcal{K}_0$ over. Now we need to find a map which maps $C_0 \times C_0 \times C_0 \times C_0$ back to $C \times C \times C \times C$ and we moreover need that this mapping is locally Lipschitz on $\mathbb{R}^4 \setminus (C_0 \times C_0 \times C_0 \times C_0)$. Note that $T^{-1}$ is not locally Lipschitz there and hence cannot be used for this.

*Step 3. Homeomorphism that maps $C_0 \times C_0 \times C_0 \times C_0$ onto $C \times C \times C \times C$.*

We will find a sequence of homeomorphisms $F_k : (-1,1)^4 \to (-1,1)^4$. We set $F_0(x) = x$ and we proceed by induction. We will give a mapping $F_1$ which stretches each cube $\tilde{Q}_v$, $v \in \mathbb{V}^1$ , homogeneously so that $F_1(\tilde{Q}_v)$ equals $Q_v$. On the annulus $\tilde{Q}'_v \setminus \tilde{Q}_v$, $F_1$ is defined to be an appropriate radial map with respect to $z_v$ and $\tilde{z}_v$ in the image in order to make $F_1$ a homeomorphism. The general step is the following: If $k > 1$, $F_k$ is defined as $F_{k-1}$ outside the union of all cubes $\tilde{Q}'_{\boldsymbol{v}}$, $\boldsymbol{v} \in \mathbb{V}^k$. Further, $F_k$ remains equal to $F_{k-1}$ at the centers of cubes $\tilde{Q}_{\boldsymbol{v}}$, $\boldsymbol{v} \in \mathbb{V}^k$. Then $F_k$ stretches each cube $\tilde{Q}_{\boldsymbol{v}}$, $\boldsymbol{v} \in \mathbb{V}^k$, homogeneously so that $F_k(\tilde{Q}_{\boldsymbol{v}})$ equals $Q_{\boldsymbol{v}}$. On the annulus $\tilde{Q}'_{\boldsymbol{v}} \setminus \tilde{Q}_{\boldsymbol{v}}$, $F_k$ is defined to be an appropriate radial map with respect to $\tilde{z}_{\boldsymbol{v}}$ in preimage and $z_{\boldsymbol{v}}$ in image to make $F_k$ a homeomorphism (see Fig. 4). Notice that the Jacobian determinant $J_{F_k}(x)$ will be strictly positive almost everywhere in $(-1,1)^4$.

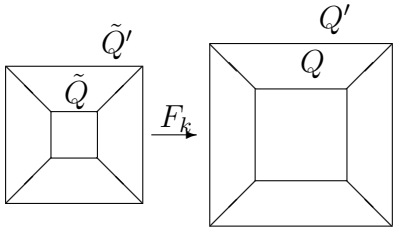



**Fig. 4.** The transformation of $\tilde{Q}' \setminus \tilde{Q}^\circ$ onto $Q' \setminus Q^\circ$ (2D projection)

In the following definition of $F_k$ we use the notation $\|x\|$ for the supremum norm of $x \in \mathbb{R}^n$. The mappings $F_k$, $k \in \mathbb{N}$, are formally defined as

$$F_k(x) = \begin{cases} F_{k-1}(x) & \text{for } x \notin \bigcup_{\boldsymbol{v} \in \mathbb{V}^k} \tilde{Q}'_{\boldsymbol{v}} \\ F_{k-1}(\tilde{z}_{\boldsymbol{v}}) + (\alpha_k \|x - \tilde{z}_{\boldsymbol{v}}\| + \beta_k) \frac{x - \tilde{z}_{\boldsymbol{v}}}{\|x - \tilde{z}_{\boldsymbol{v}}\|} & \text{for } x \in \tilde{Q}'_{\boldsymbol{v}} \setminus \tilde{Q}_{\boldsymbol{v}},\ \boldsymbol{v} \in \mathbb{V}^k \\ F_{k-1}(\tilde{z}_{\boldsymbol{v}}) + \frac{r_k}{\tilde{r}_k}(x - \tilde{z}_{\boldsymbol{v}}) & \text{for } x \in \tilde{Q}_{\boldsymbol{v}},\ \boldsymbol{v} \in \mathbb{V}^k \end{cases} \tag{4.2}$$

where the constants $\alpha_k$ and $\beta_k$ are given by

$$\alpha_k \tilde{r}_k + \beta_k = r_k \text{ and } \alpha_k \tfrac{\tilde{r}_{k-1}}{2} + \beta_k = \tfrac{r_{k-1}}{2}. \tag{4.3}$$

It is not difficult to find out that each $F_k$ is a homeomorphism and maps

$$\bigcup_{\boldsymbol{v} \in \mathbb{V}^k} \tilde{Q}_{\boldsymbol{v}} \text{ onto } \bigcup_{\boldsymbol{v} \in \mathbb{V}^k} Q_{\boldsymbol{v}}.$$

The limit $F(x) = \lim_{k \to \infty} F_k(x)$ is clearly one to one and continuous and therefore a homeomorphism. Moreover, it is easy to see that $F$ is differentiable almost everywhere, absolutely continuous on almost all lines parallel to coordinate axes and maps $C_0 \times C_0 \times C_0 \times C_0$ onto $C \times C \times C \times C$ and its pointwise Jacobian is positive a.e. (recall that $\mathcal{L}^1(C_0) = 0$).

Fix $k_0 \in \mathbb{N}$. We claim that the mapping $F$ is Lipschitz on $(-1,1)^4 \setminus \bigcup_{\boldsymbol{v} \in \mathbb{V}^{k_0}} \tilde{Q}'_{\boldsymbol{v}}$ (the Lipschitz constant of course depends on the fixed $k_0$). This is in fact easy to see as

the mapping is given by simple formula (4.2) on each $\tilde{Q}'_{\boldsymbol{v}} \setminus \tilde{Q}_{\boldsymbol{v}}$ for every $\boldsymbol{v} \in \mathbb{V}^k$ and $k \leq k_0$. Analogously to [20, Lemma 2.1 and proof of Theorem 4.10] we can estimate

$$|DF(x)| \sim \max\Big\{\frac{r_k}{\tilde{r}_k}, \alpha_k\Big\}$$

for every $x \in \tilde{Q}'_{\boldsymbol{v}} \setminus \tilde{Q}_{\boldsymbol{v}}$ and $\boldsymbol{v} \in \mathbb{V}^k$. It follows that $F$ is locally Lipschitz on $(-1,1)^4 \setminus (C_0 \times C_0 \times C_0 \times C_0)$ (with constant bounded by a constant multiple of $\max_{1\leq k\leq k_0} \max\{\frac{\tilde{r}_k}{r_k}, \alpha_k\}$ on $(-1,1)^4 \setminus \bigcup_{\boldsymbol{v}\in\mathbb{V}^{k_0}} Q(z_{\boldsymbol{v}}, r_{k_0})$).

*Step 4. ACL condition and estimate of derivative $Df_1$.*

Let us consider the homeomorphism $f_1 := F \circ h \circ T$, where $h$ is from Theorem 3.4 and $F$ and $T$ are defined above. We would like to verify the ACL condition to show that $f_1 \in W^{1,1}$ by Theorem 2.1.

Let us consider a line segment $L := (-1,1) \times \{z\}$ and its image. For $z \notin C \times C \times C$ we know that $T(L)$ is a line segment parallel to $x_1$ axis and its image by $h$ is Lipschitz curve that does not intersect $C_0 \times C_0 \times C_0 \times C_0$ and hence its image $f_1(L)$ is a Lipschitz curve and thus it is absolutely continuous curve and it has bounded length. For $z \in C \times C \times C$ we know that $T(L)$ is a line segment parallel to $x_1$ axis intersecting $C_0 \times C_0 \times C_0 \times C_0$ and by (3.2) its image by $h$ is the same segment $T(L)$. We claim that $f_1(L) = F(T(L))$ has bounded length. Indeed, it intersects each $Q'_{\boldsymbol{v}} \setminus Q_{\boldsymbol{v}}$, $\boldsymbol{v} \in \mathbb{V}^k$, in two line segments of length at most $2\sqrt{4}(\frac{r_{k-1}}{2} - r_k)$ and the sum of those lengths over all $k$ and all $2^k$ rectangular annuli it intersects is at most $2\sqrt{4}$ and thus the length of $f_1(L) \setminus \mathcal{C}$ is bounded. Moreover, it follows from the construction of $f_1$ that $f_1(L \cap \mathcal{C}) = L \cap \mathcal{C}$ and hence the length of $f_1(L)$ is bounded by $2\sqrt{4} + 2 = 6$. Using these facts it is also easy to see that $f_1(L)$ is absolutely continuous curve for $z \in C \times C \times C$. Similar result is true for segments parallel to other coordinate axes.

It remains to estimate $\int |Df_1|$. For simplicity we estimate only the derivative with respect to the first coordinate $|D_1 f_1|$ as the other derivatives can be estimated analogously. Let us denote by $\tau(x_1, x_2, x_3, x_4) = [x_2, x_3, x_4]$ the projection to the last three coordinates. We set

$$\mathcal{A}_k := (-1,1)^3 \setminus \bigcup_{\boldsymbol{v}\in\mathbb{V}^k} Q(\tau(z_{\boldsymbol{v}}), r_k) \ .$$

It is easy to see that

$$\bigcup_{k\in\mathbb{N}} \mathcal{A}_k = (-1,1)^3 \setminus C \times C \times C \ .$$

Since $h$ is Lipschitz and $F$ is locally Lipschitz on $\mathbb{R}^4 \setminus (C \times C \times C \times C)$ it is easy to see that

$$l_i := \sup\big\{\mathcal{H}^1(f_1(L)) : L = (-1,1) \times \{z\} \text{ for some } z \in \mathcal{A}_i\big\} < \infty.$$

As the integral of the derivative corresponds to the length of the image we obtain that the derivative with respect to the first coordinate can be estimated as

$$\begin{aligned}\int_{(-1,1)^4} |D_1 f_1| &= \int_{(-1,1)\times C\times C\times C} |D_1 f_1| + \int_{(-1,1)^4\setminus((-1,1)\times C\times C\times C)} |D_1 f_1| \\ &\leq \mathcal{H}^3(C \times C \times C)6 + \sum_i \mathcal{H}^3\big(\mathcal{A}_i \setminus \mathcal{A}_{i-1}\big) l_i \ ,\end{aligned}$$

but the last sum maybe infinite as we have no control over $l_i$. We will fix this by considering a bigger Cantor type set of positive measure.

*Step 5. Change from $\mathcal{K}$ to bigger $\hat{\mathcal{K}}$.*

We can clearly choose a sequence $\hat{a}_k$ such that $\hat{a}_0 = 1$, $\hat{a}_k > \hat{a}_{k+1}$, $\lim_{k\to\infty} \hat{a}_k = \hat{a} > 0$ and

$$\sum_{i=1}^{\infty} (\hat{a}_{i-1}^3 - \hat{a}_i^3) l_i < \infty \ , \tag{4.4}$$

since $l_i$ are fixed. Similarly as in Step 1 and 2 we set

$$\hat{r}_k = \hat{a}_k 2^{-k} \ .$$

As before we define

$$\hat{z}_{\boldsymbol{v}} = \hat{z}_{\boldsymbol{w}} + \frac{1}{2}\hat{r}_{k-1} v_k = z_0 + \frac{1}{2}\sum_{j=1}^{k} \hat{r}_{j-1} v_j,$$

$$\hat{Q}'_{\boldsymbol{v}} = Q(\hat{z}_{\boldsymbol{v}}, \tfrac{\hat{r}_{k-1}}{2}) \ \text{ and } \ \hat{Q}_{\boldsymbol{v}} = Q(\hat{z}_{\boldsymbol{v}}, \hat{r}_k).$$

It is not difficult to find out that the resulting Cantor set

$$\bigcap_{k=1}^{\infty} \bigcup_{\boldsymbol{v}\in\mathbb{V}^k} \hat{Q}_{\boldsymbol{v}} =: \hat{\mathcal{C}} = \hat{C} \times \hat{C} \times \hat{C} \times \hat{C}$$

satisfies $\mathcal{L}^4(\hat{\mathcal{C}}) > 0$ since

$$\mathcal{L}^4\Big( \bigcup_{\boldsymbol{v}\in\mathbb{V}^k} \hat{Q}_{\boldsymbol{v}}\Big) = 2^{4k}(2\hat{a}_k 2^{-k})^4 \overset{k\to\infty}{\to} 16\hat{a}^4.$$

Similarly to Step 3 there is a homeomorphism $s : \mathbb{R} \to \mathbb{R}$ which maps $\hat{C}$ onto $C$. It is just enough to consider a mapping which maps $\pi(\hat{Q}'_{\boldsymbol{v}}) \setminus \pi(\hat{Q}_{\boldsymbol{v}})$ onto $\pi(Q'_{\boldsymbol{v}}) \setminus \pi(Q_{\boldsymbol{v}})$ linearly for each $\boldsymbol{v} \in \mathbb{V}^k$ and $k \in \mathbb{N}$. Then the following mapping

$$S(x_1, x_2, x_3, x_4) = [s(x_1), s(x_2), s(x_3), s(x_4)]$$

is a homeomorphism, maps $\hat{\mathcal{K}}$ onto $\mathcal{K}$ and moreover it maps line segments parallel to coordinate axes to segments parallel to coordinate axes.

We now consider a homeomorphism $f := f_1 \circ S = F \circ h \circ T \circ S$. As both $S$ and $T$ map line segments parallel to coordinate axes to segments parallel to coordinate axes it is easy to see that analogously to the previous step this $f$ is absolutely continuous on almost all lines parallel to coordinate axes. We set

$$\hat{\mathcal{A}}_k := (-1,1)^3 \setminus \bigcup_{\boldsymbol{v}\in\mathbb{V}^k} Q(\tau(\hat{z}_{\boldsymbol{v}}), \hat{r}_k)$$

and observe that lines $(-1,1)\times\{\hat{z}\}$ for $\hat{z} \in \hat{\mathcal{A}}_k$ are mapped by $S$ to lines $(-1,1)\times\{z\}$ for some $z \in \mathcal{A}_k$. It follows that analogously to the previous step we have

$$\begin{aligned} \int_{(-1,1)^4} |D_1 f| &= \int_{(-1,1)\times\hat{C}\times\hat{C}\times\hat{C}} |D_1 f| + \int_{(-1,1)^4\setminus((-1,1)\times\hat{C}\times\hat{C}\times\hat{C})} |D_1 f| \\ &\le \mathcal{H}^3(\hat{C}\times\hat{C}\times\hat{C})6 + \sum_i \mathcal{H}^3\big(\hat{\mathcal{A}}_i \setminus \hat{\mathcal{A}}_{i-1}\big) l_i \ . \end{aligned} \tag{4.5}$$

In the projection of $\bigcup_{\boldsymbol{v}\in\mathbb{V}^k} Q(\tau(\hat{z}_{\boldsymbol{v}}), \hat{r}_k)$ we have $2^{3k}$ cubes ($2^k$ cubes in the $x_1$ direction have the same projection) and hence

$$\mathcal{H}^3\Big( \bigcup_{\boldsymbol{v}\in\mathbb{V}^k} Q(\tau(\hat{z}_{\boldsymbol{v}}), \hat{r}_k)\Big) = 2^{3k}(2\hat{r}_k)^3 = 8\hat{a}_k^3 \ .$$

It follows that

$$\mathcal{H}^3\big(\hat{\mathcal{A}}_k \setminus \hat{\mathcal{A}}_{k-1}\big) = 8(\hat{a}_{k-1}^3 - \hat{a}_k^3)$$

and hence (4.4), (4.5) and Theorem 2.1 imply that $f \in W^{1,1}$.

Outside of $\hat{\mathcal{K}}$ our mapping is a sense preserving homeomorphism that is locally Lipschitz and hence $J_f \geq 0$ there (see e.g. [21]). We can also require that $J_f$ is not identically zero there (otherwise by [19] $f$ does not satisfy Lusin $(N)$ condition there which cannot happen for locally Lipschitz map) and hence $\mathcal{L}^4(\{x : \ J_f(x) > 0\}) > 0$. On the other hand $f$ is a homeomorphism that maps $\hat{\mathcal{C}}$ onto $\mathcal{C}$ in a regular way

$$f(x_1, x_2, x_3, x_4) = [-s(x_1), s(x_2), s(x_3), s(x_4)] \text{ for every } x \in \hat{\mathcal{C}} \ .$$

It is well-known fact that each Sobolev function is approximatively differentiable a.e. Since

$$\lim_{k\to\infty} \frac{r_k}{\hat{r}_k} = \lim_{k\to\infty} \frac{a_k}{\hat{a}_k} = \frac{1}{2\hat{a}} > 0$$

it is not difficult to find out that at almost every point of $\hat{\mathcal{C}}$ we have

$$Df(x) = \begin{pmatrix} -\frac{1}{2\hat{a}} & 0 & 0 & 0 \\ 0 & \frac{1}{2\hat{a}} & 0 & 0 \\ 0 & 0 & \frac{1}{2\hat{a}} & 0 \\ 0 & 0 & 0 & \frac{1}{2\hat{a}} \end{pmatrix}$$

and thus $\mathcal{L}^4(\{x : \ J_f(x) < 0\}) > 0$. □

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